

CFD Modeling of Air Flow around the Ahmed Car Model using ANSYS Fluent

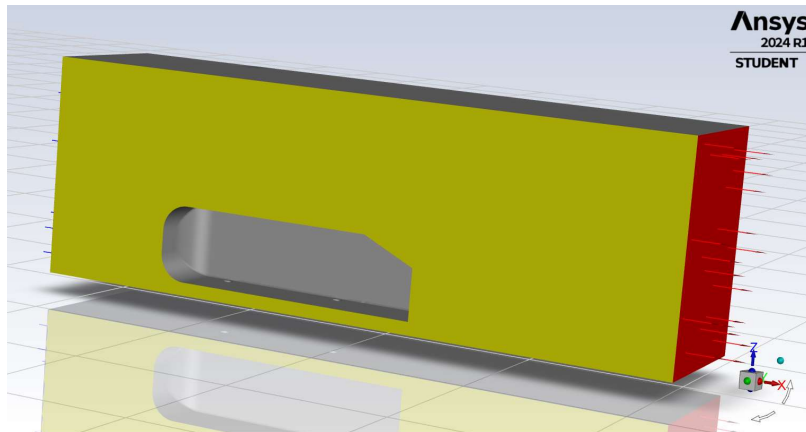
Zachery Taylor
Department of Mechanical Engineering
University of West Florida

Problem Description

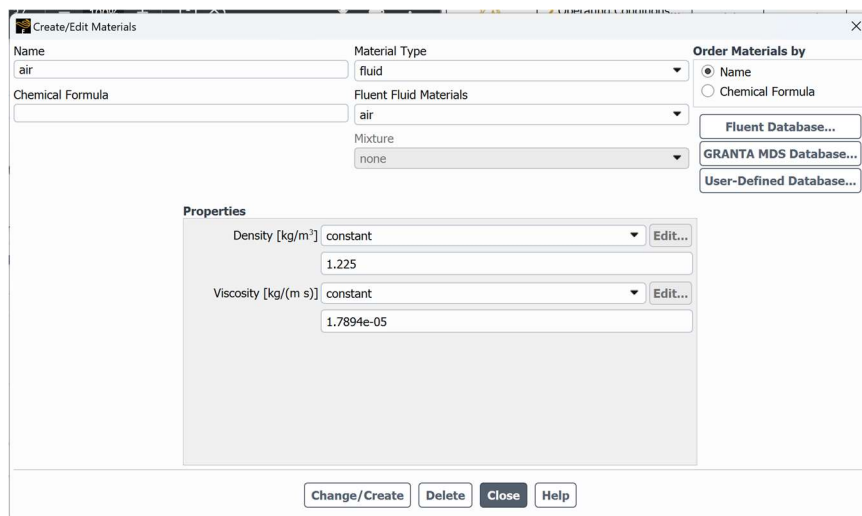
Geometry

Modeling Ahmed Car Model; looking to see how air reacts around it.

For dimensions, please refer to the Ahmed Car Model provided file. Model was not built by the person completing this report.



Fluid Properties



Flow Conditions

Inlet: Velocity = 40 m/s
 Turbulent Intensity = 5%
 Turbulent Viscosity Ratio = 10

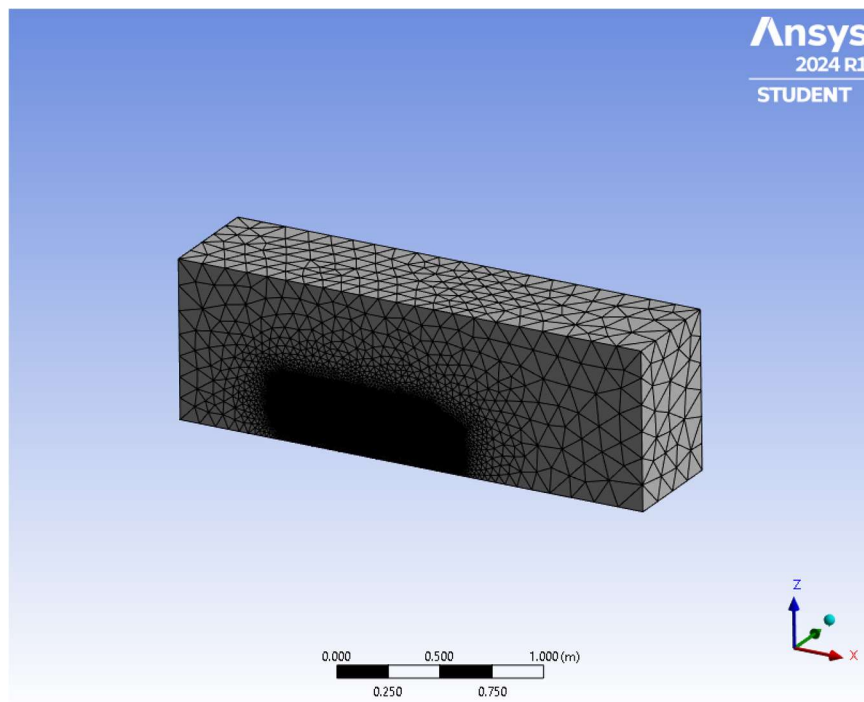
Outlet: Backflow Turbulent Intensity = 5%
 Backflow Turbulent Viscosity

Method

Mesh

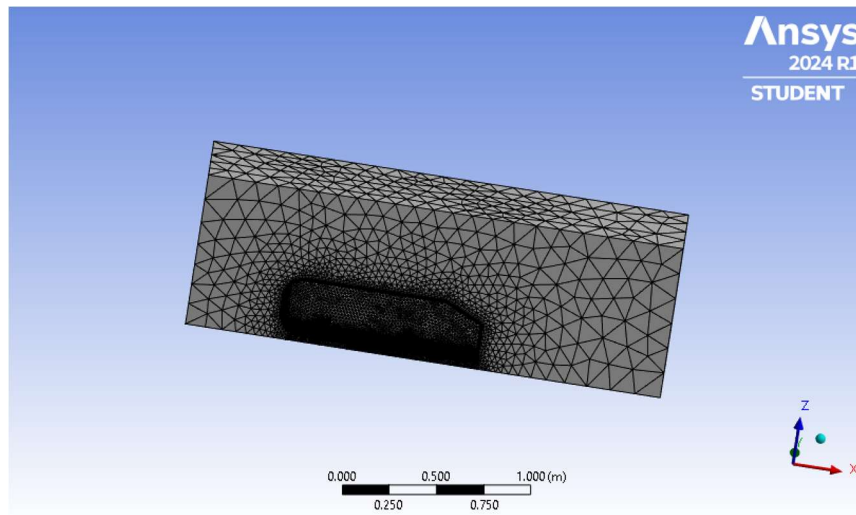
0.008m Mesh sizing

Number of Cells: 436355



0.016m Mesh sizing

Number of Cells: 202907



Equations

Viscous Model

Model

- ☐ Inviscid
- ☐ Laminar
- ☐ Spalart-Allmaras (1 eqn)
- ☒ k-epsilon (2 eqn)
- ☐ k-omega (2 eqn)
- ☐ Transition k-kl-omega (3 eqn)
- ☐ Transition SST (4 eqn)
- ☐ Reynolds Stress (7 eqn)
- ☐ Scale-Adaptive Simulation (SAS)
- ☐ Detached Eddy Simulation (DES)
- ☐ Large Eddy Simulation (LES)

k-epsilon Model

- ☒ Standard
- ☐ RNG
- ☐ Realizable

Near-Wall Treatment

- ☐ Standard Wall Functions
- ☐ Scalable Wall Functions
- ☐ Non-Equilibrium Wall Functions
- ☒ Enhanced Wall Treatment
- ☐ Menter-Lechner
- ☐ User-Defined Wall Functions

Enhanced Wall Treatment Options

- ☐ Pressure Gradient Effects

Options

- ☐ Curvature Correction
- ☐ Production Kato-Launder
- ☐ Production Limiter

Model Constants

Cmu: 0.09

C1-Epsilon: 1.44

C2-Epsilon: 1.92

TKE Prandtl Number: 1

TDR Prandtl Number: 1.3

User-Defined Functions

Turbulent Viscosity: none

Prandtl Numbers

TKE Prandtl Number: none

TDR Prandtl Number: none

OK Cancel Help

Task Page

Solution Controls

Pseudo Time Explicit Relaxation Factors

Pressure

0.25

Momentum

0.25

Density

1

Body Forces

1

Turbulent Kinetic Energy

0.75

Turbulent Dissipation Rate

0.75

Turbulent Viscosity

1

Spatial Discretization Methods

Solution Methods

Pressure-Velocity Coupling

Scheme

Coupled

Flux Type

Rhie-Chow: momentum based

☒ Auto Select

Spatial Discretization

Gradient

Least Squares Cell Based

Pressure

Second Order

Momentum

Second Order Upwind

Turbulent Kinetic Energy

Second Order Upwind

Turbulent Dissipation Rate

Second Order Upwind

Pseudo Time Method

Global Time Step

Residual Values

Residual Monitors

Options

- ☒ Print to Console
- ☒ Plot
- Curves... Axes...
- Iterations to Plot: 1000
- Iterations to Store: 1000

Equations

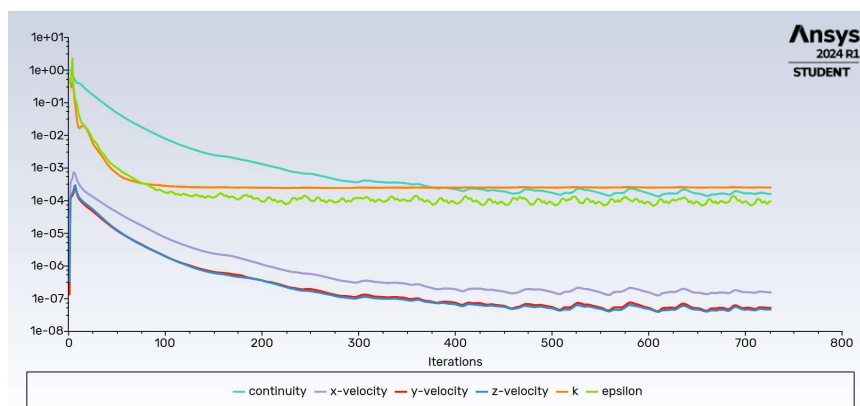
Residual	Monitor	Check Convergence	Absolute Criteria
continuity	☒	☒	1e-15
x-velocity	☒	☒	1e-15
y-velocity	☒	☒	1e-15
z-velocity	☒	☒	1e-15
k	☒	☒	1e-15
epsilon	☒	☒	1e-15

Convergence Conditions...

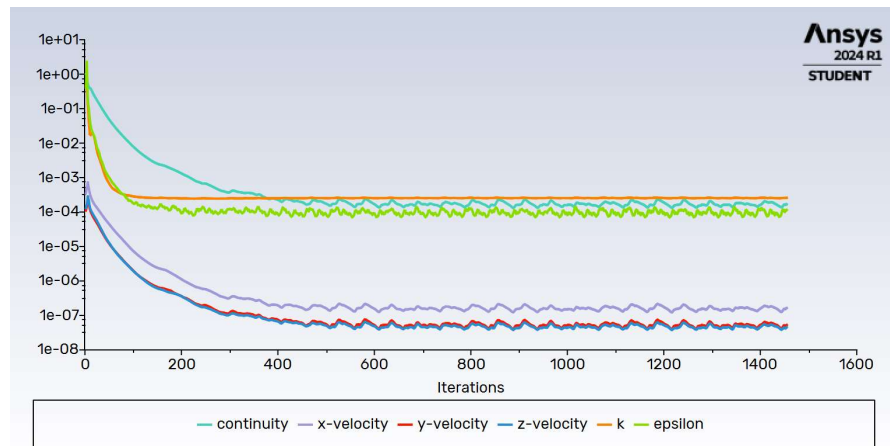
☐ Show Advanced Options

OK Plot Cancel Help

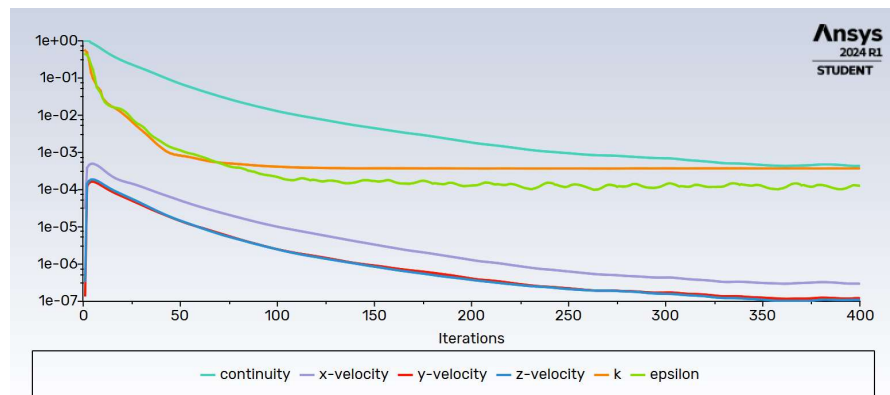
0.008m Mesh Scaled Residuals plot after 725 iterations



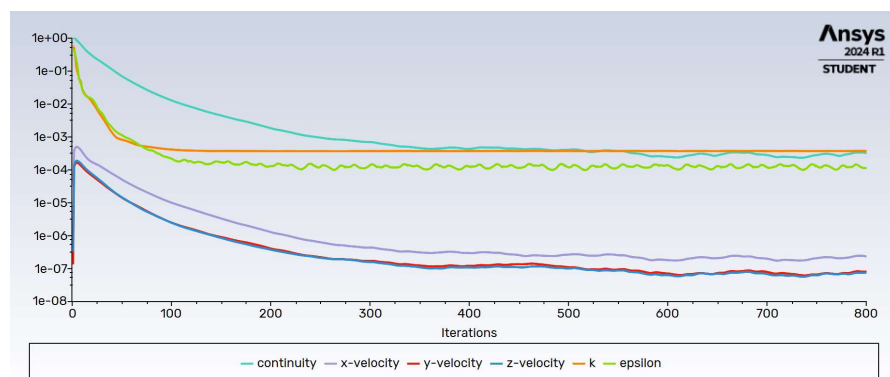
0.008m Mesh Scaled Residuals plot after 1450 iterations



0.016m Mesh Scaled Residuals plot after 400 iterations



0.016m Mesh Scaled Residuals plot after 800 iterations



Results

Drag Report Definition

Name: report-def-0

Options

☐ Per Zone

Average Over(Iterations): 1

Force Vector

X: 1 Y: 0 Z: 0

Report Files [0/0]

Report Plots [0/0]

Create

☒ Report File

☒ Report Plot

Frequency: 1

☐ Print to Console

☐ Create Output Parameter

Report Output Type

☐ Drag Coefficient

☒ Drag Force

Zones: Filter Text

car

road

wall

OK Compute Cancel Help

Lift Report Definition

Name: report-def-1

Options

☐ Per Zone

Average Over(Iterations): 1

Force Vector

X: 0 Y: 0 Z: 1

Report Files [0/1]

report-def-0-rfile

Report Plots [0/1]

report-def-0-rplot

Create

☒ Report File

☒ Report Plot

Frequency: 1

☐ Print to Console

☐ Create Output Parameter

Report Output Type

☐ Lift Coefficient

☒ Lift Force

Zones: Filter Text

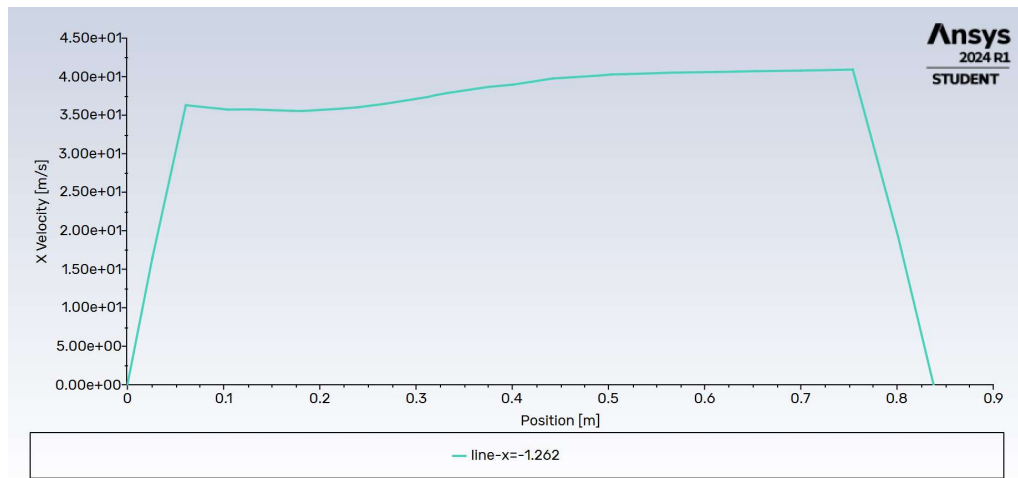
car

road

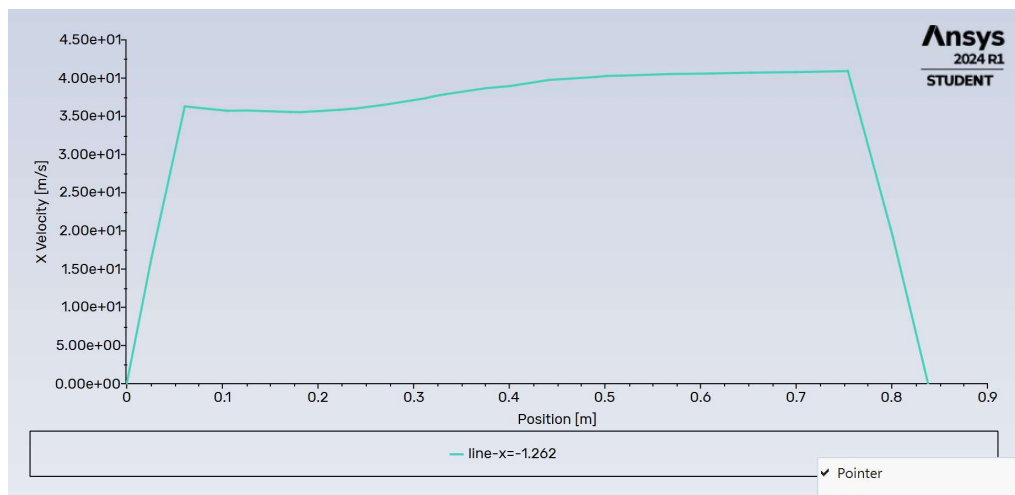
wall

OK Compute Cancel Help

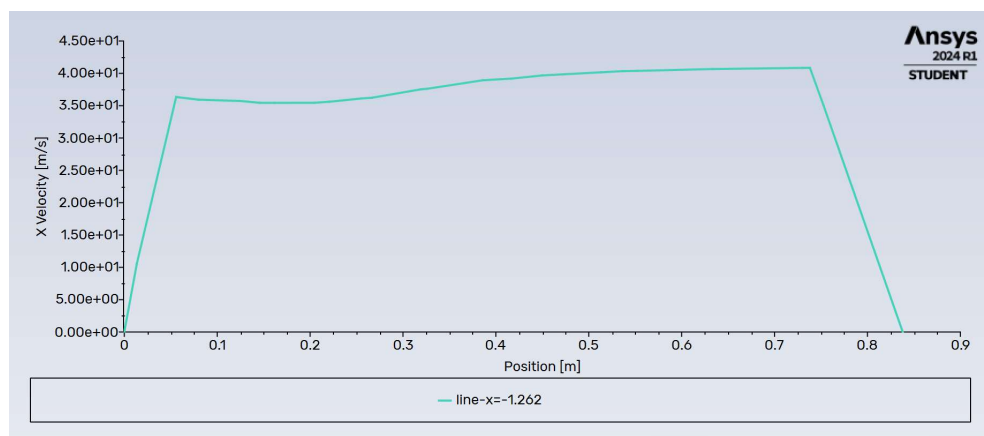
0.008m Mesh X-Velocity plot at $x=-1.262$, Plot Direction (0,0,1) after 725 iterations:



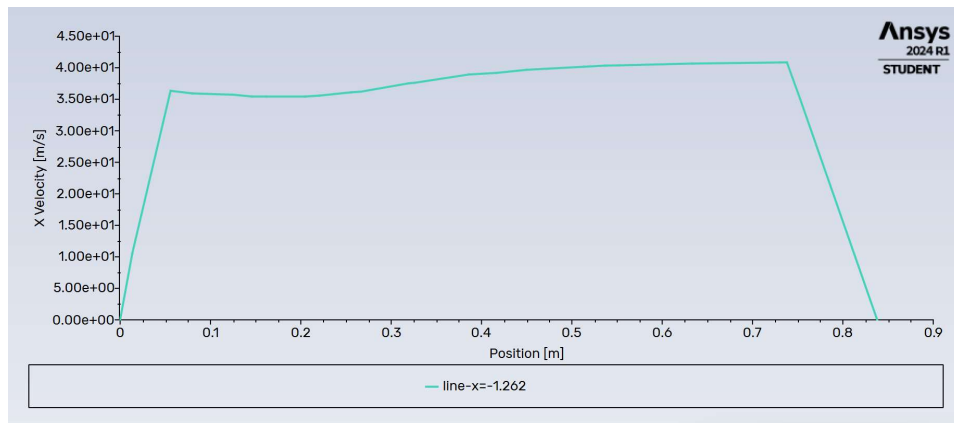
0.008m Mesh X-Velocity plot at $x=-1.262$, Plot Direction (0,0,1) after 1450 iterations:



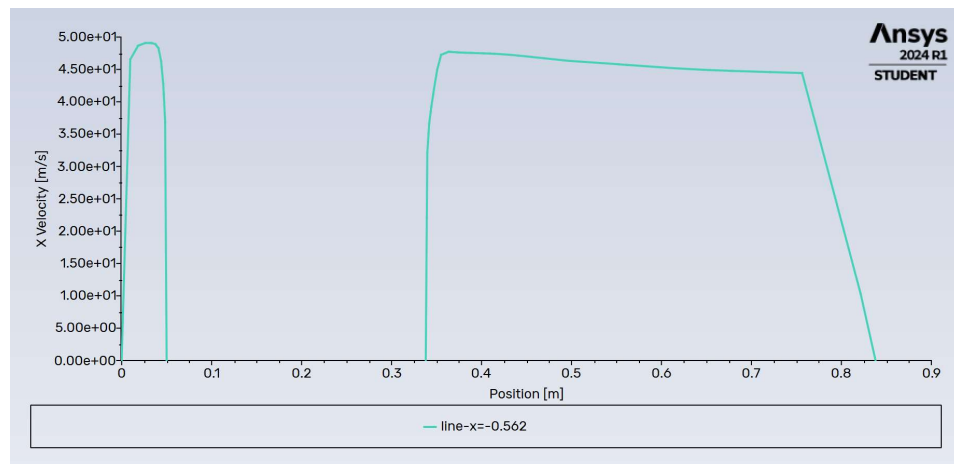
0.016m Mesh X-Velocity plot at $x=-1.262$, Plot Direction (0,0,1) after 400 iterations:



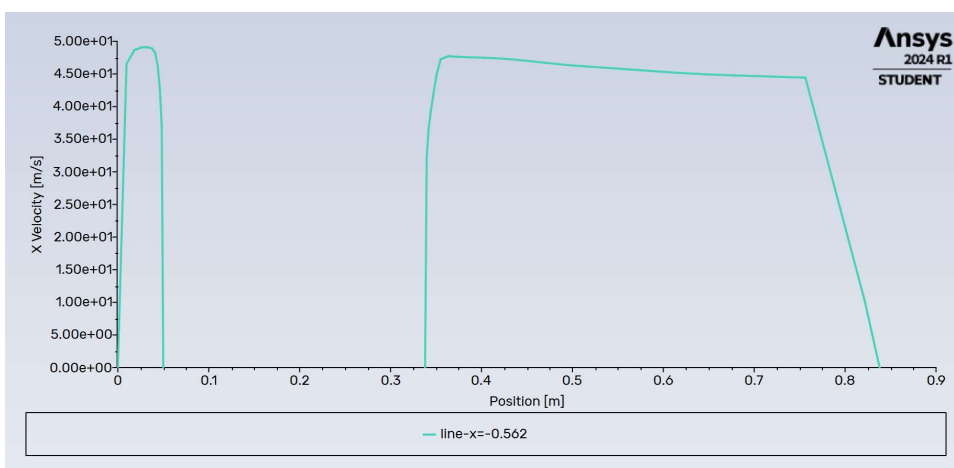
0.016m Mesh X-Velocity plot at $x=-1.262$, Plot Direction (0,0,1) after 800 iterations:



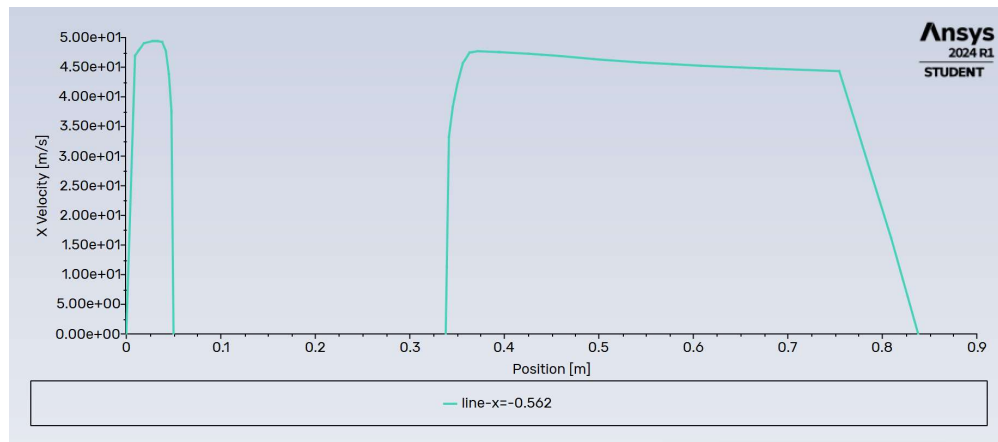
0.008m Mesh X-Velocity plot at $x=-0.562$, Plot Direction (0,0,1) after 725 iterations:



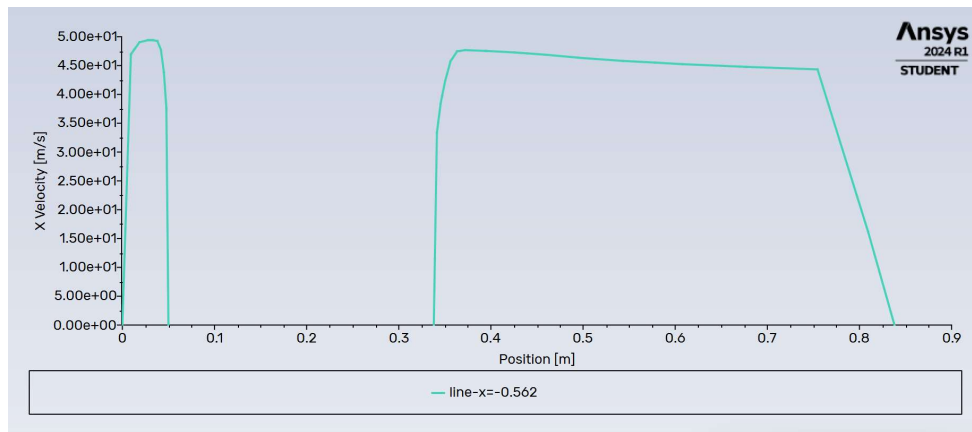
0.008m Mesh X-Velocity plot at $x=-0.562$, Plot Direction (0,0,1) after 1450 iterations:



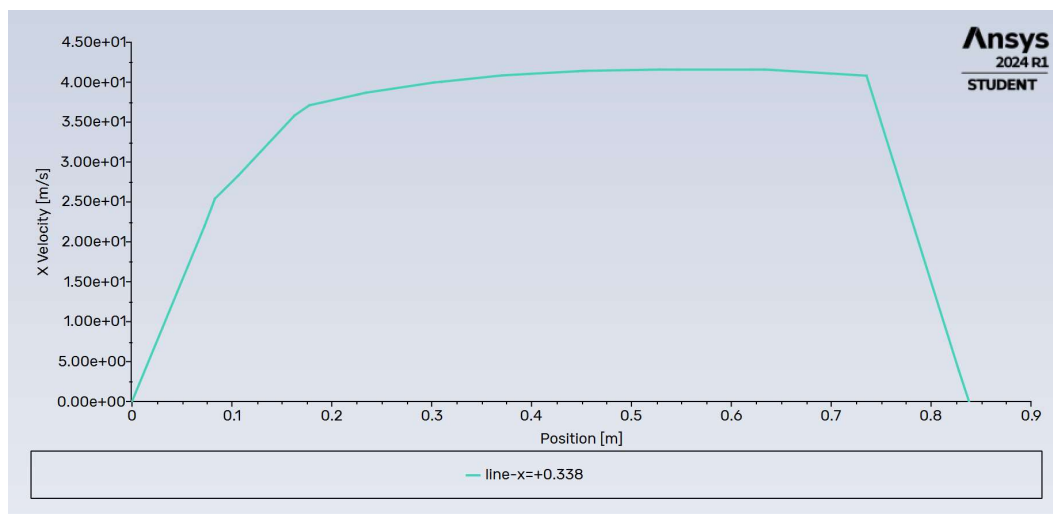
0.016m Mesh X-Velocity plot at $x=-0.562$, Plot Direction (0,0,1) after 400 iterations:



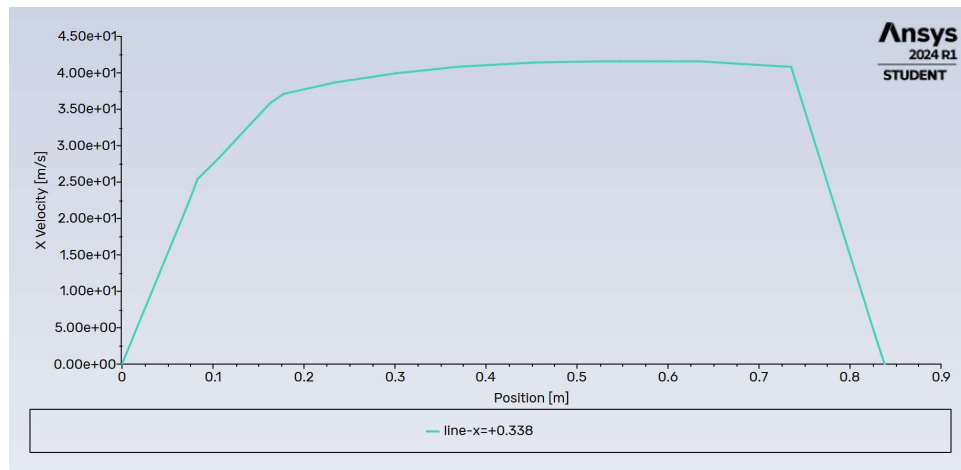
0.016m Mesh X-Velocity plot at $x=-0.562$, Plot Direction (0,0,1) after 800 iterations:



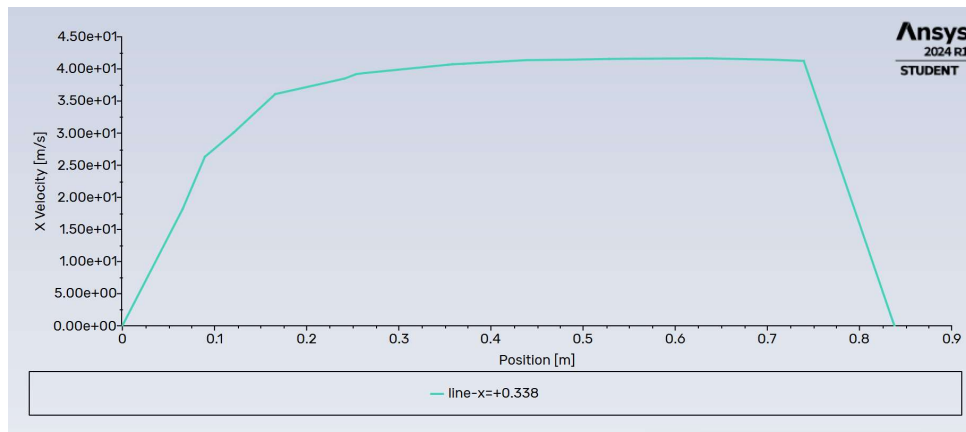
0.008m Mesh X-Velocity plot at $x=+0.338$, Plot Direction (0,0,1) after 725 iterations:



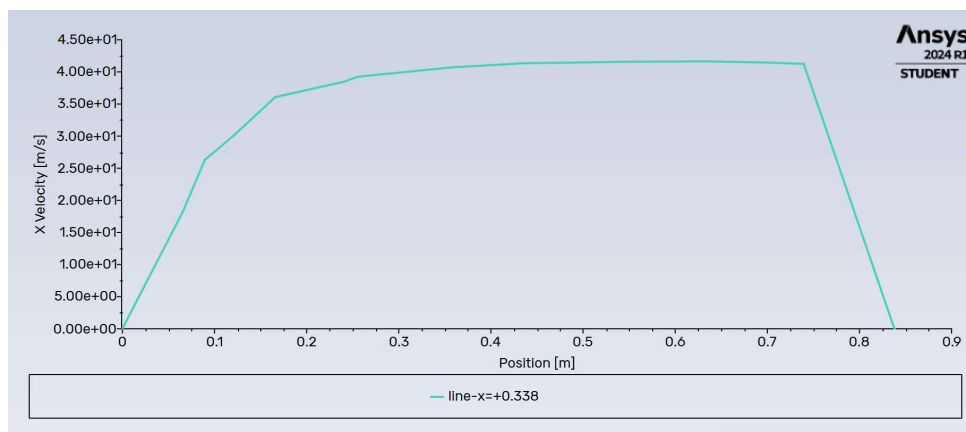
0.008m Mesh X-Velocity plot at $x=+0.338$, Plot Direction (0,0,1) after 1450 iterations:



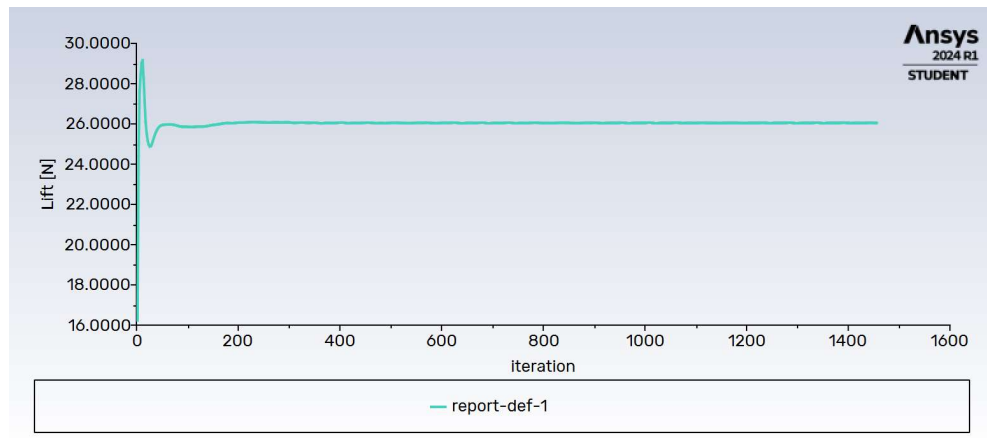
0.016m Mesh X-Velocity plot at $x=+0.338$, Plot Direction (0,0,1) after 400 iterations:



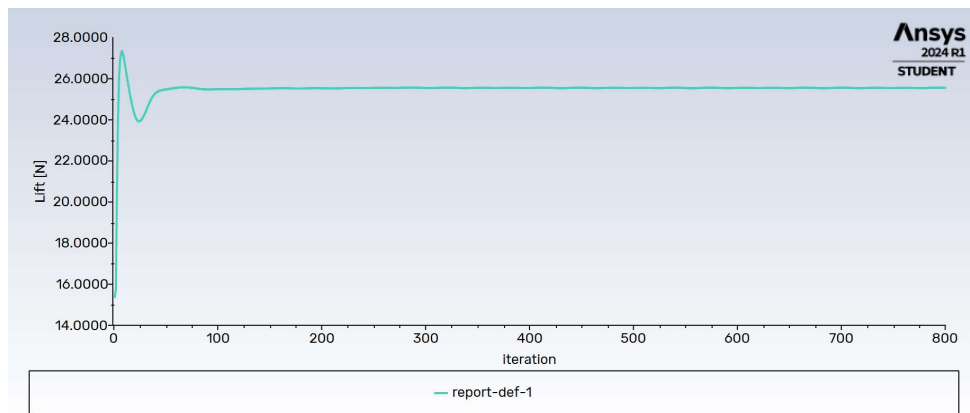
0.016m Mesh X-Velocity plot at $x=+0.338$, Plot Direction (0,0,1) after 800 iterations:



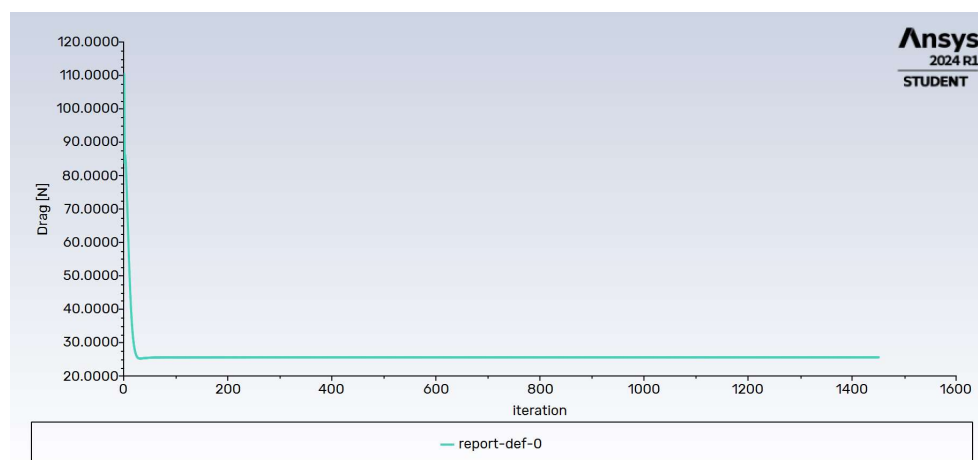
0.008m Mesh Lift plot after 1450 iterations:



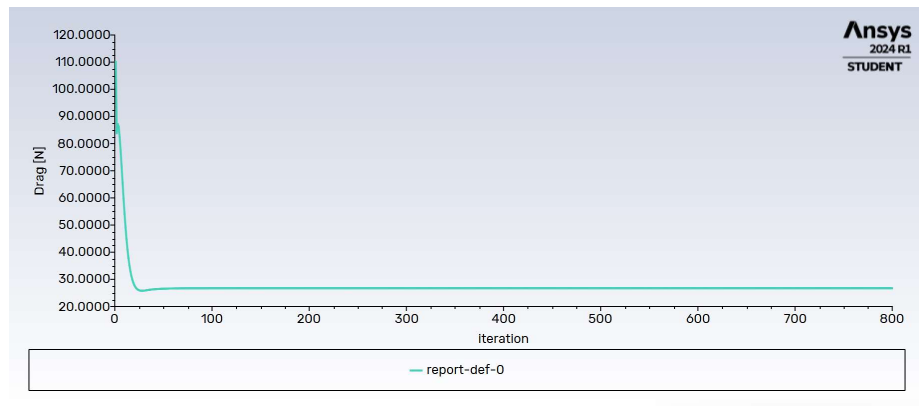
0.016m Mesh Lift plot after 800 iterations:



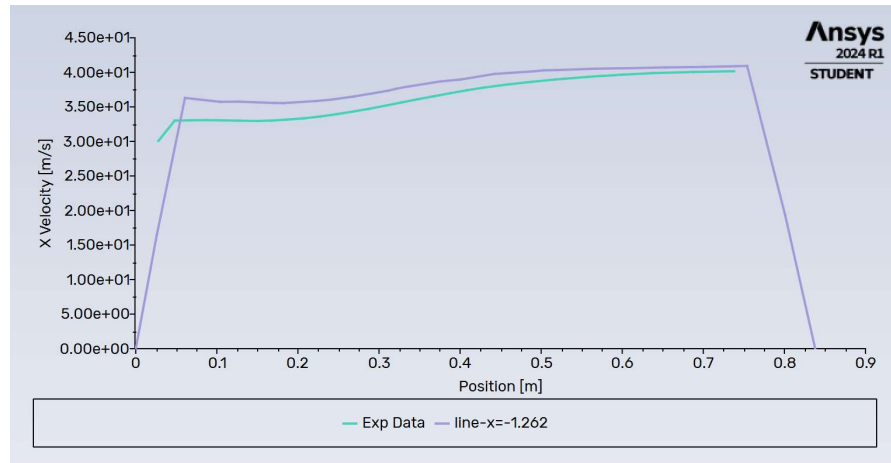
0.008m Mesh Drag plot after 1450 iterations:



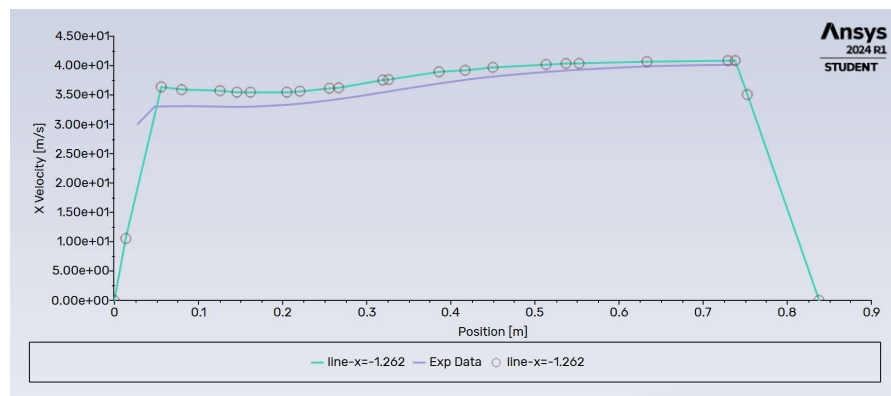
0.016m Mesh Drag plot after 800 iterations:



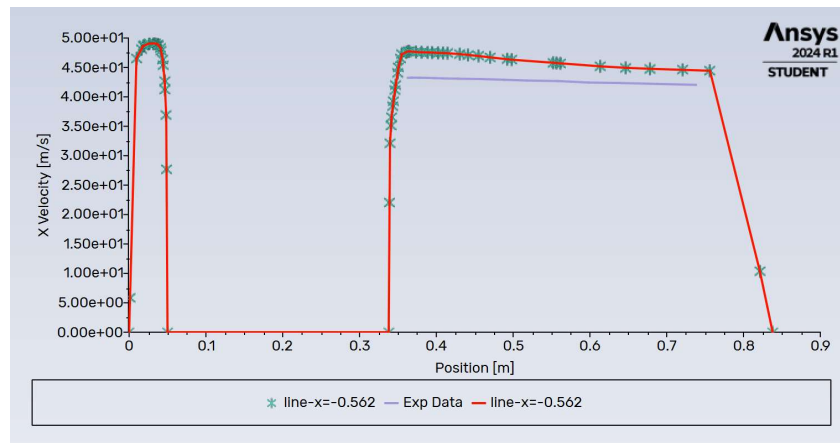
0.008m Mesh X-Velocity plot at $x=-1.262$, Plot Direction (0,0,1) after 725 iterations vs. 1450 iterations vs. Experimental Data



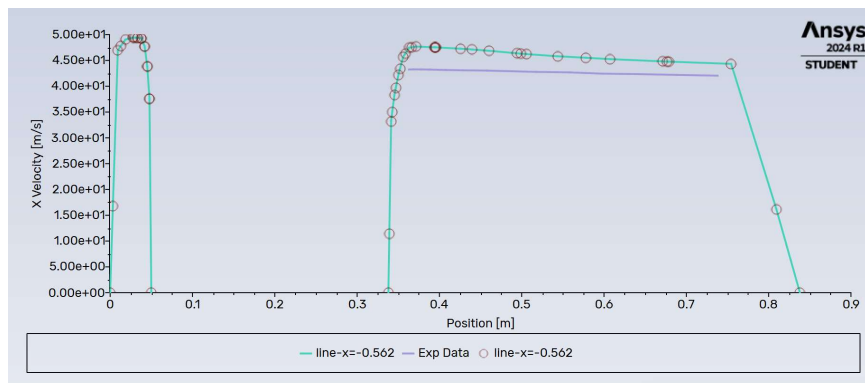
0.016m Mesh X-Velocity plot at $x=-1.262$, Plot Direction (0,0,1) after 400 iterations vs. 800 iterations vs. Experimental Data



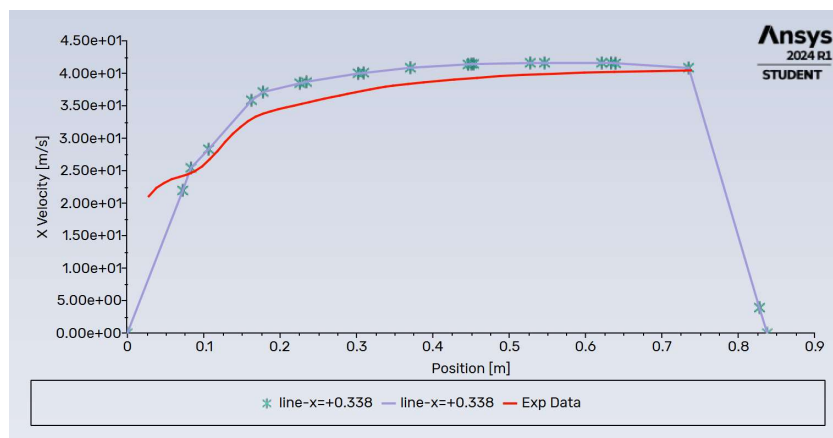
0.008m Mesh X-Velocity plot at $x=-0.562$, Plot Direction (0,0,1) after 725 iterations vs. 1450 iterations vs. Experimental Data



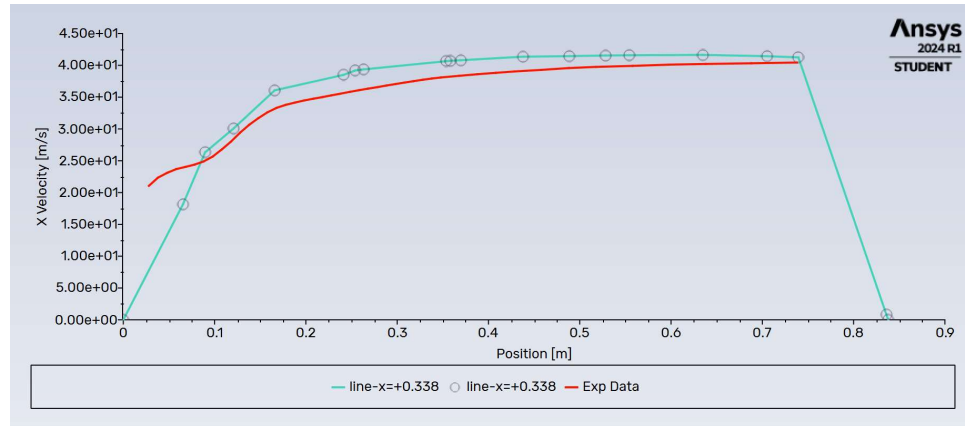
0.016m Mesh X-Velocity plot at $x=-0.562$, Plot Direction (0,0,1) after 400 iterations vs. 800 iterations vs. Experimental Data



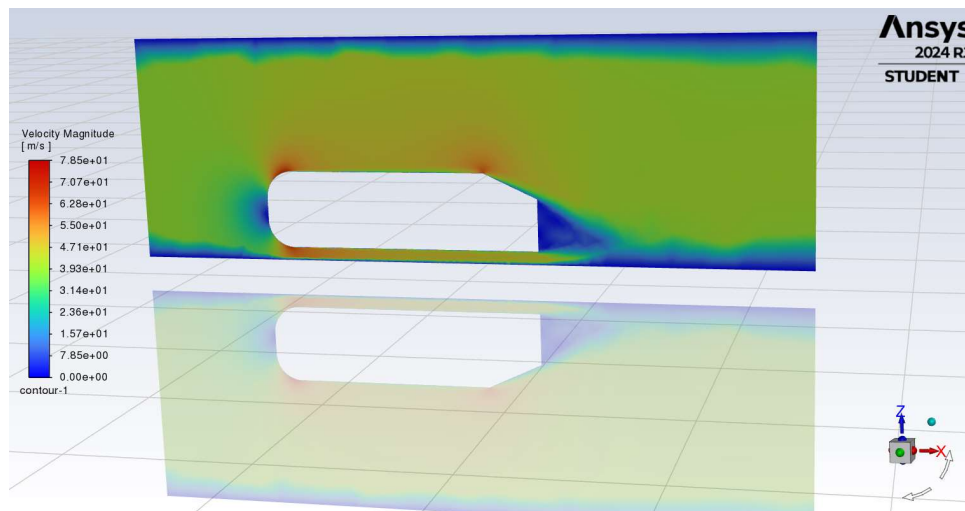
0.008m Mesh X-Velocity plot at $x=+0.338$, Plot Direction (0,0,1) after 725 iterations vs. 1450 iterations vs. Experimental Data



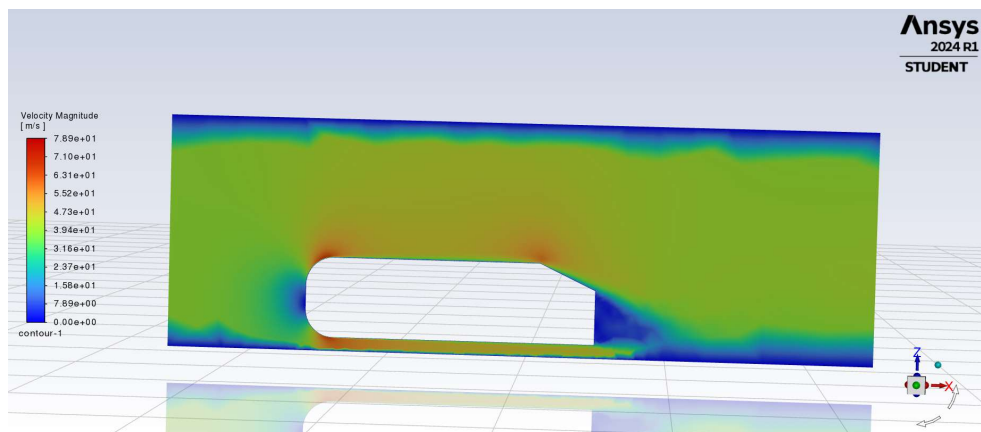
0.016m Mesh X-Velocity plot at $x=+0.338$, Plot Direction (0,0,1) after 400 iterations vs. 800 iterations vs. Experimental Data



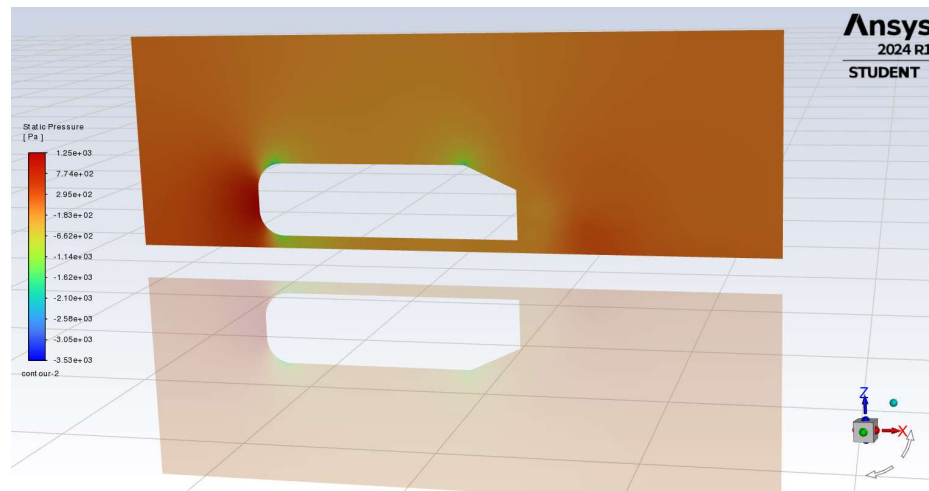
0.008m Mesh Velocity Magnitude Contour Plot



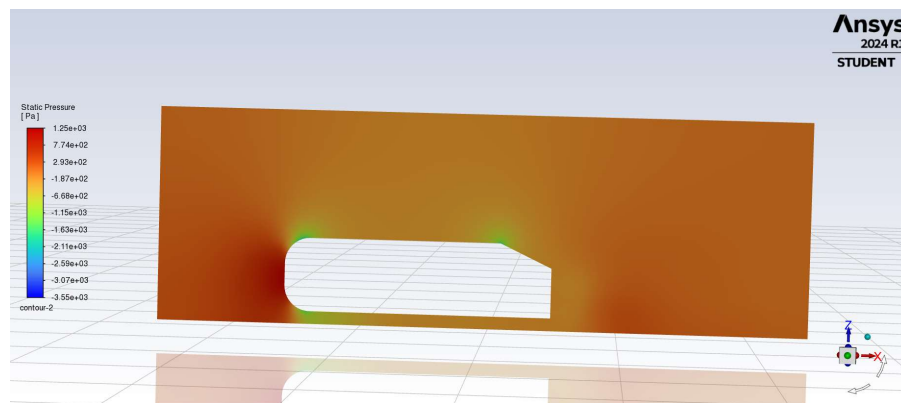
0.016m Mesh Velocity Magnitude Contour Plot



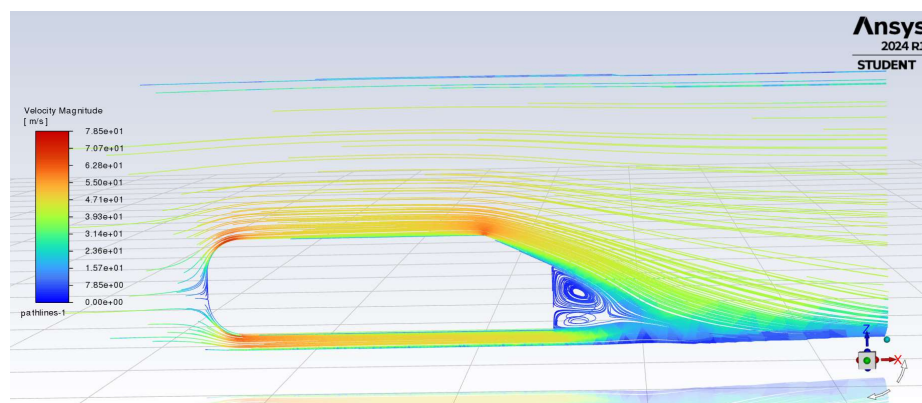
0.008m Mesh Static Pressure Contour Plot



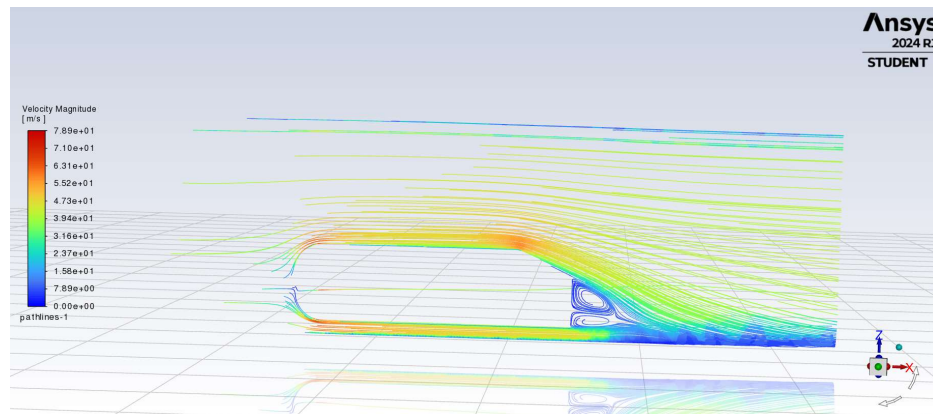
0.016m Mesh Static Pressure Contour Plot



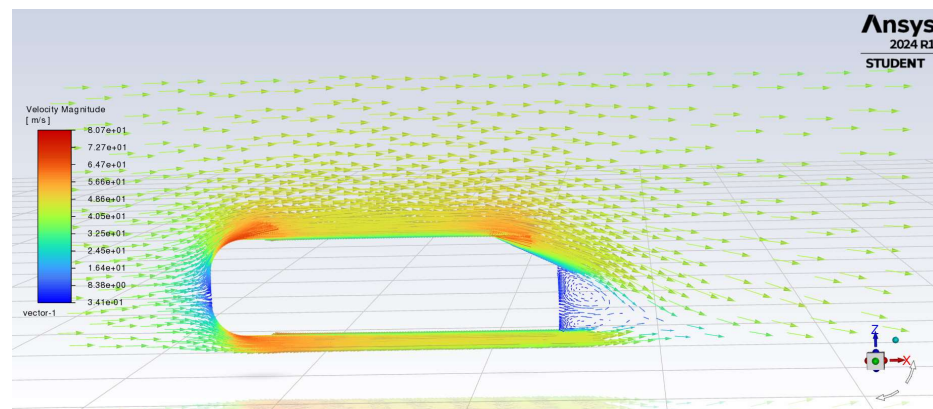
0.008m Mesh Pathline Plot



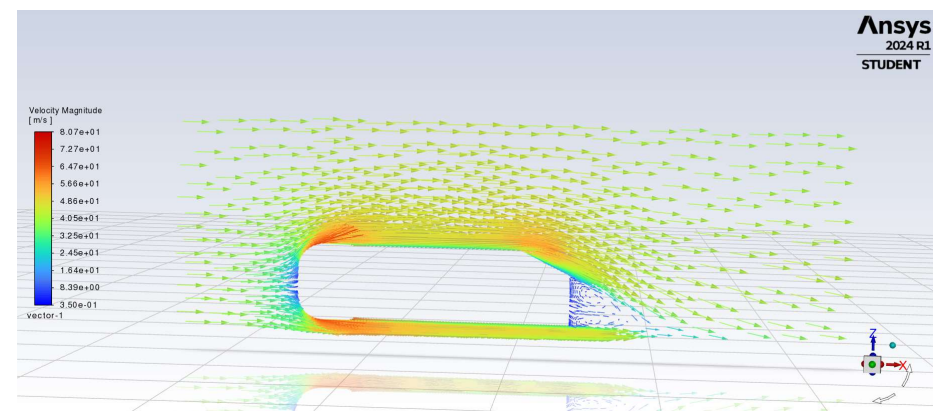
0.016m Mesh Pathline Plot



0.008m Mesh Velocity Vector Plot

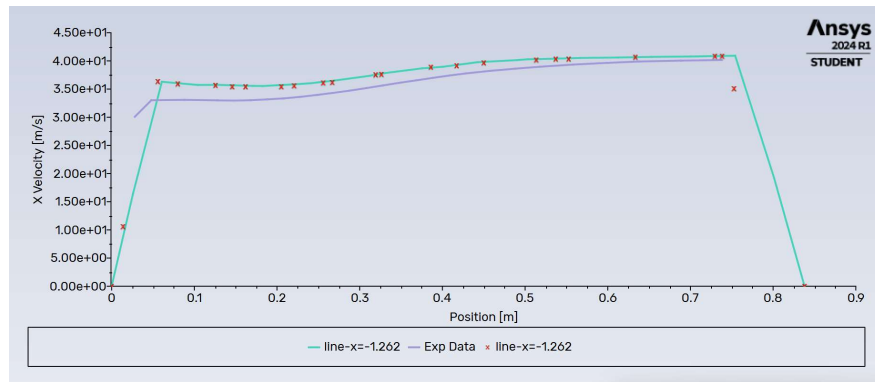


0.016m Mesh Velocity Vector Plot

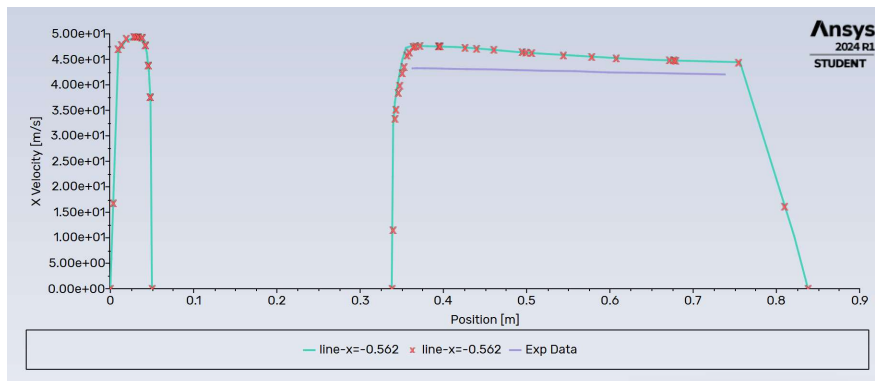


Proving Mesh Independence

X-Velocity plot at $x=-1.262$, Plot Direction (0,0,1) 0.008m Mesh after 1450 iterations vs 0.016 Mesh after 800 iterations vs. Experimental Data



X-Velocity plot at $x=-0.562$, Plot Direction (0,0,1) 0.008m Mesh after 1450 iterations vs 0.016 Mesh after 800 iterations vs. Experimental Data



X-Velocity plot at $x=+0.338$, Plot Direction (0,0,1) 0.008m Mesh after 1450 iterations vs 0.016 Mesh after 800 iterations vs. Experimental Data

